

Lab Sheet 8 (CST-108)
Part-A (I/O – C and UNIX)

Note: These are practice questions. No need to submit output or reason.

1. Study the output of the following C program.

```
int main(){
    printf("u");
    write(STDOUT_FILENO, "m", 1);
    printf("d\n");
}
```

2. How many different outputs are possible (1, 2 or many) if the following program is executed multiple times.

```
static void charatime(char *);

int main(void){
    pid_t pid;

    pid = fork();
    if (pid == 0) charatime("output from child\n");
    else charatime("output from parent\n");
    exit(0);
}

static void charatime(char *str){
    char *ptr;
    int c;

    setbuf(stdout, NULL);
    for(ptr = str; c = *ptr++; )
        putc(c, stdout);
}
```

Part-B (BIO package - File and Socket BIOs)

1. Create a blank file with name "Sample.txt". Now write a C program to do the following:
 - (i) Use fopen function to open the file Sample.txt for reading and writing.
 - (ii) Create a buffered file BIO object and associate it with FILE * returned by fopen.
 - (iii) Write string "Hello World" to this file BIO.
 - (iv) Use BIO_read to read from the BIO and display contents of the on the screen.
 - (v) Now use function fgets to read from the file Sample.txt directly and display contents of the on the screen.
 - (vi) Destroy the BIO after using it.

2. Repeat exercise 1 for unbuffered file BIO.
3. Write a C program to do the following:
 - (i) Use BIO_new_file function to create a file BIO. Use filename Sample.bin.
 - (ii) Next create a buffering filter BIO and a cipher filter BIO.
 - (iii) Use EVP_aes_256_cbc for encryption and decryption.
 - (iv) Assemble the BIO chain to be in the order cipher-buffer-file
 - (v) Encrypt string "Hello World!" using this chain and write into the file Sample.bin.
4. Write a C program to decrypt and read the contents of the file Sample.bin using the BIO chain of cipher, buffer and the file. Display the decrypted character string on the screen.
5. Re-write echo client and echo server (written in lab sheet 2 Q4) using socket BIO. Client should use following functions (in addition to other functions) to establish connection:

```
BIO_new(BIO_s_connect());  
BIO_set_conn_hostname();  
BIO_set_conn_port();  
BIO_do_connect();
```

Server should use following functions (in addition to other functions) for binding address / port and to listen the requests from clients:

```
BIO_new_accept();  
BIO_do_accept();
```

Server should create a new process (not thread) to handle each request from the clients.